Ogbonna Arinze

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SUMMARY

Senior Full-Stack Engineer with 8 years of professional experience building high-impact web applications across healthcare, fintech, civic tech, and developer-facing platforms. Deep expertise in **React, Next.js, TypeScript**, and **Node.js**, with production experience in **GraphQL** APIs, **CI/CD pipeline engineering** with GitHub Actions, and **performance optimization**. Proven track record of shipping end-to-end features with full ownership in small, high-velocity teams, from architecting monorepo platforms managing 6 applications to building automated testing and deployment pipelines. Passionate about developer tooling, code quality, and building products that engineering teams rely on.

CORE COMPETENCIES

- **Full-Stack Development:** React, Next.js, TypeScript, Node.js, NestJS, GraphQL, REST APIs, Vue.js, Redux, Zustand, and Tailwind CSS.
- **CI/CD & Integrations:** GitHub Actions, automated testing pipelines, Docker, Docker Compose, Nx monorepo orchestration, and deployment automation.
- **Performance Engineering:** Core Web Vitals, code splitting, lazy loading, CDN strategies, Lighthouse, bundle analysis (webpack-bundle-analyzer, rollup-plugin-visualizer), and render profiling.
- **Backend & Data:** NestJS, Node.js REST/GraphQL APIs, PostgreSQL (pgvector), MongoDB, Prisma, JWT authentication, RBAC, and SSE streaming.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA with automated reporting.
- **AI & Data Pipelines:** RAG systems, OpenAI API, LangChain, Vercel AI SDK, vector search (pgvector, Chroma), document ingestion pipelines, and multi-agent workflows.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Vercel, Docker, Traefik, Infisical, GitHub Pages, and multi-stage Dockerfile deployments.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and product-minded development in small teams.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)
Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension** with **TypeScript**, owning the feature end-to-end from architecture through production deployment.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, with **GraphQL** for data layer integration, enabling users to query their data in real time via a conversational interface.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, measuring and validating improvements via **Lighthouse** and Chrome DevTools performance profiling.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating test reporting with **Allure Reports** on GitHub Pages, and configuring Slack notifications for build quality visibility.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through structured code reviews.

Descasio — Lagos, Nigeria (Hybrid)
Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application, directly improving Core Web Vitals and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and more confident deployments.

- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, processing thousands to millions of daily transactions and boosting settlement rates by **80%**, demonstrating ability to deliver impactful features in a small engineering team.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios**, handling complex real-time data flows.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface, reducing latency for live transaction monitoring.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**, enabling stakeholders to monitor transaction performance metrics.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, owning the full backend implementation.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance, code reviews, and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Built a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline, powered by a **hybrid RAG pipeline** combining **pgvector** similarity search + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking across 708K+ vector embeddings.
- Built the NestJS document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing 700+ documents (959 files) across 37 states with zero ingestion errors.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** reverse proxy routing.
- Implemented **SSE streaming** with 15s keepalive heartbeats, **PostgreSQL 16** with **Prisma v7**, phone OTP authentication, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*